

5(a)(i)	<u>Mycobacterium tuberculosis</u> or <u>Mycobacterium bovis</u> ;	1
5(a)(ii)	<i>max1 if no reference to, infected / uninfected</i> airborne droplets, breathed / sneezed / AW, out by infected person ; breathed in / inhaled / inspired, by uninfected person ; or <i>in context of M. bovis only</i> contaminated meat / unpasteurised milk, from infected, cattle / cows ; ingested by uninfected person ;	2
5(b)	<i>max 4 if no ref. to, (mature) B-lymphocytes / plasma cells, producing / secreting, antibodies</i> <i>any five from:</i> <i>ref. to</i> fusion of lysosomes releasing hydrolytic enzymes ; intracellular digestion / described ; e.g. hydrolysis of, (named) cell wall components / proteins / nucleic acids / lipids <i>ref. to</i> processing antigens to present on, cell surface membrane / cell surface ; A macrophage becomes an, antigen presenting cell / APC ; antigen, recognition / binding, by, B-lymphocytes / T-lymphocytes ; detail ; e.g. clonal selection / have receptors complementary to antigen / specificity of, B-lymphocytes / T- lymphocytes to (non-self) antigen ; A immunoglobulin / antibody as receptor for B-lymphocytes clonal, expansion / proliferation ; A described e.g. divides by mitosis to form a clone of cells B-lymphocytes, mature to / form, plasma cells that secrete antibodies ; T-helper cells release cytokines ; Cytokine action ; e.g. stimulates, humoral / B-lymphocyte, response	5

Question	Answer	Marks
5(c)	<i>any three from:</i> long-term immunity ; B and T memory cells remain in the body for, a long time / a lifetime ; memory cells for secondary (immune) response ; secondary response is faster / more effective (than primary immune response) ; (so) higher concentration antibodies / quicker production antibodies ; <i>Idea of</i> more (specific) memory cells in the body than (specific) lymphocytes before primary response ; increase chance that pathogen encountered more rapidly (to mount a response) ; AVP ; e.g. pathogen destroyed before symptoms of illness are caused	3
5(d)	<i>any three from:</i> <i>Mycobacterium</i> / TB bacterium, is an intracellular parasite / lives inside host cells ; antibodies have no effect on bacteria inside cells ; only when bacteria are in the, plasma / tissue fluid ; <i>idea that</i> vaccination is not a global occurrence ; e.g. not part of vaccination programme in many countries only some countries have as part of their vaccination programme BCG is not very effective (for adults) / some people do not respond to BCG ; BCG does not have the same effectiveness across the world ; many countries, have no herd immunity / do have enough people that are immune, to prevent spread ; ref. to civil disturbance / AW, collapse of health services ; AVP ; <i>idea that</i> travellers / tourists / immigrants / displaced people, may be infected with TB and transmit it to host population (that does not have, a vaccination programme / herd immunity)	3